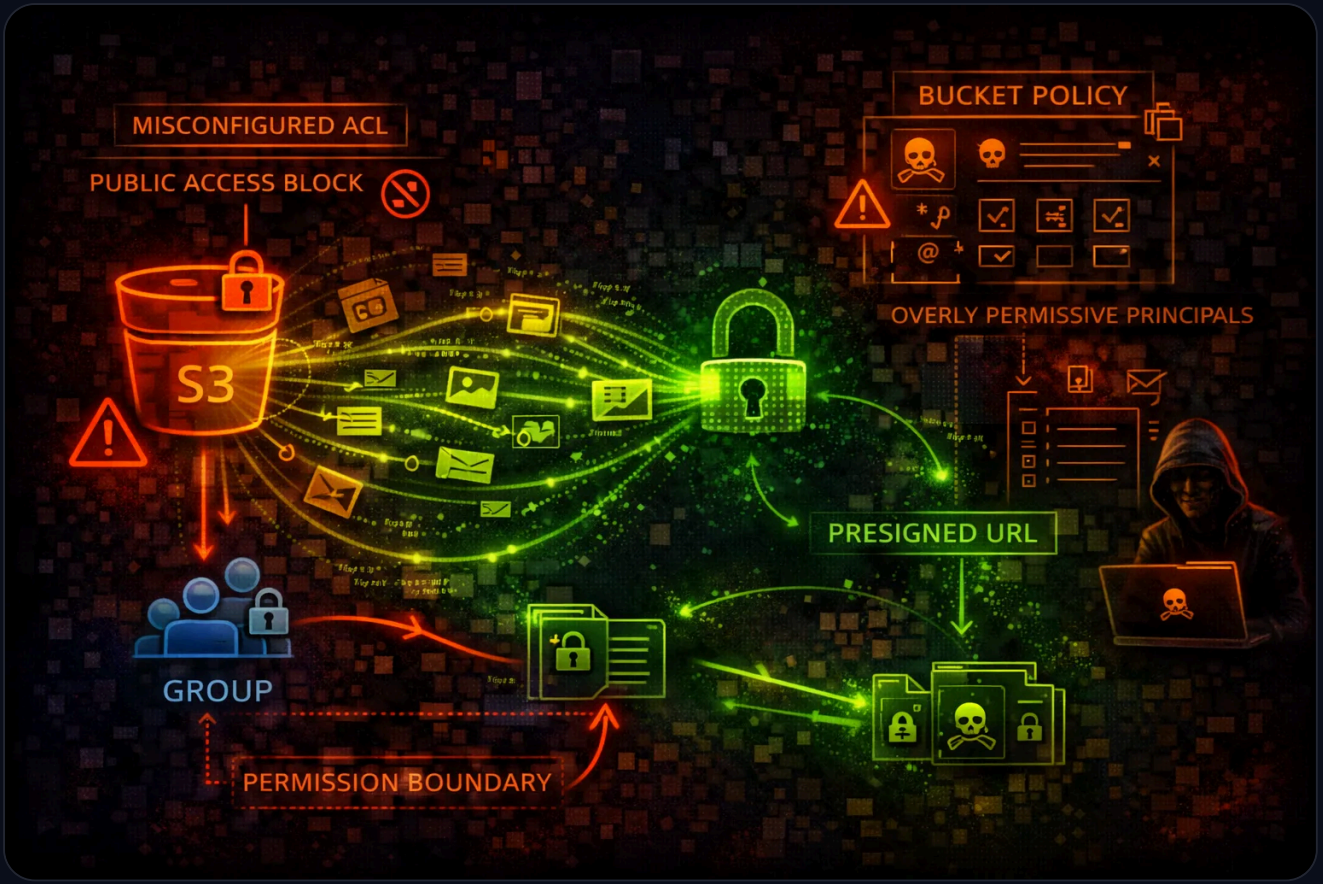


AWS S3 Security

STORAGE

Simple Storage Service (S3) provides object storage with bucket policies and ACLs controlling access. S3 is the #1 source of cloud data breaches due to misconfigured public buckets and weak policies.

CRITICAL	Global	Objects	Policies
Risk Level	Namespace	Up to 5TB Each	+ ACLs

Service Overview

Bucket Access Controls

S3 uses bucket policies (resource-based) and ACLs (legacy) to control access. Block Public Access settings provide account-level and bucket-level guardrails against accidental exposure.

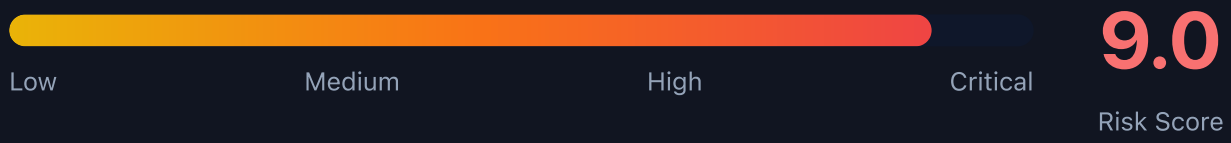
Attack note: Public buckets are discovered within hours using automated scanning tools

Data Access Methods

Objects can be accessed via API, CLI, Console, or pre-signed URLs. Pre-signed URLs grant temporary access without requiring AWS credentials - often leaked or misconfigured.

Attack note: SSRF vulnerabilities can access private buckets via EC2 instance roles

Security Risk Assessment



S3 misconfigurations are responsible for the majority of cloud data breaches. Public buckets, overly permissive policies, and lack of encryption expose sensitive data to the internet.

⚔️ Attack Vectors

Public Exposure

- Public buckets with sensitive data
- Bucket policy allows Principal: *
- ACL grants public-read or public-read-write
- Block Public Access disabled
- Website hosting exposes directory listing

Access Abuse

- Pre-signed URL leaked or shared
- SSRF to access private buckets
- Cross-account access too permissive
- Versioning disabled - data destruction
- Replication to attacker bucket

⚠️ Misconfigurations

Policy Issues

- Principal: * without conditions
- s3:* action allowed broadly
- No VPC endpoint restriction
- Missing aws:SecureTransport condition
- Cross-account access too wide

Security Settings

- Block Public Access disabled
- No server-side encryption
- Object Lock not enabled
- Access logging not configured
- MFA Delete not enabled

🔍 Enumeration

List All Buckets

```
aws s3 ls
```

List Bucket Contents

```
aws s3 ls s3://bucket-name --re
cursive
```

Get Bucket Policy

```
aws s3api get-bucket-policy --b
ucket NAME
```

Get Bucket ACL

```
aws s3api get-bucket-acl --buck
et NAME
```

Check Public Access Block

```
aws s3api get-public-access-blo
ck \\
  --bucket NAME
```

📤 Data Exfiltration

Bulk Download

- aws s3 sync - download entire bucket
- List and download specific prefixes
- Access all object versions
- Download via pre-signed URLs
- Stream objects to external storage

Replication Abuse

- Configure cross-region replication
- Replicate to attacker-controlled bucket
- Batch operations for bulk export
- S3 Access Points for data access
- Use Object Lambda to exfil

Gold Mine: Backup buckets often contain database dumps, logs with credentials, and configuration files.

📝 Object Manipulation

Write Access Abuse

- Overwrite website index.html
- Upload malicious JavaScript
- Replace application configs
- Plant backdoors in deployments
- Modify log files to hide tracks

Destruction

- Delete objects and versions
- Ransomware - encrypt objects
- Modify bucket lifecycle rules
- Delete bucket (if empty)
- Disable versioning first

🛡️ Detection

Logging & Monitoring

- S3 Server Access Logs
- CloudTrail data events
- CloudWatch metrics
- Macie sensitive data findings
- GuardDuty S3 protection

Indicators of Compromise

- Unusual GetObject patterns
- Mass downloads from new IPs
- Public access configuration changes
- Bucket policy modifications
- Cross-account access from unknown

💻 Exploitation Commands

Download Entire Bucket

```
aws s3 sync s3://target-bucket ./loot
```

Upload Malicious File

```
aws s3 cp malicious.html s3://bucket/index.html
```

Make Bucket Public

```
aws s3api put-bucket-acl \\
  --bucket NAME --acl public-read
```

Check Anonymous Access

```
aws s3 ls s3://bucket-name --no-sign-request
```

Generate Pre-signed URL

```
aws s3 presign s3://bucket/secret.txt \\
  --expires-in 604800
```

Delete All Versions

📜 Policy Examples

✕ Dangerous - Public Bucket

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": "*",
    "Action": "s3:*",
    "Resource": ["arn:aws:s3:::bucket/*"]
  }]
}
```

Anyone on the internet can read, write, and delete all objects

✓ Secure - VPC Restricted

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": {"AWS": "arn:aws:iam::123456789012:role/AppRole"},
    "Action": ["s3:GetObject"],
    "Resource": "arn:aws:s3:::bucket/data/*",
    "Condition": {
      "StringEquals": {"aws:SourceVpc": "vpc-12345"}
    }
  }]
}
```

Only specific role can access, and only from within VPC

✕ Dangerous - Any AWS Account

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": {"AWS": "*"},
    "Action": ["s3:GetObject", "s3:PutObject"],
    "Resource": "arn:aws:s3:::bucket/*"
  }]
}
```

Any authenticated AWS user can read/write - not just your account

✓ Secure - HTTPS Only

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Deny",
    "Principal": "*",
    "Action": "s3:*",
    "Resource": "arn:aws:s3:::bucket/*",
    "Condition": {
      "Bool": {"aws:SecureTransport": "false"}
    }
  }]
}
```

Deny all requests not using HTTPS - prevents data interception

Defense Recommendations

Block Public Access

Enable at account and bucket level - this is the most important setting.

```
aws s3api put-public-access-block --bucket NAME \\
  --public-access-block-configuration \\
  BlockPublicAcls=true,IgnorePublicAcls=true,\\
BlockPublicPolicy=true,RestrictPublicBuckets=true
```

Enable Default Encryption

SSE-S3, SSE-KMS, or SSE-C for all objects automatically.

```
aws s3api put-bucket-encryption --bucket NAME \\
  --server-side-encryption-configuration ...
```

Enable Versioning

Protect against accidental deletion and ransomware attacks.

```
aws s3api put-bucket-versioning --bucket NAME \\
  --versioning-configuration Status=Enabled
```

Enable Access Logging

Track all access to bucket for security monitoring.

```
aws s3api put-bucket-logging --bucket NAME \\
  --bucket-logging-status ...
```

Use VPC Endpoints

Restrict bucket access to within your VPC only.

```
"Condition": {"StringEquals": \\
  {"aws:SourceVpc": "vpc-xxx"}}
```

Enable Macie

Discover and protect sensitive data automatically.

```
aws macie2 enable-macie && \\
aws macie2 create-classification-job ...
```


Always obtain proper authorization before testing